

Enrolment Number										
------------------	--	--	--	--	--	--	--	--	--	--

Total No. of printed pages = 03

Monsoon, 2025
B.Tech 3rd Semester Examination
Fluid Mechanics
Course Code: BCE23305T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Part A

Answer ALL questions

Multiple Choice (10 x 1 mark = 10 marks)

- Q.1 (i) The specific gravity of a liquid has 10 x 1 = 10
- (a) the same unit as that of mass density
 - (b) the same unit as that of weight density
 - (c) the same unit as that of specific volume
 - (d) no unit
- (ii) The property by virtue of which liquid opposes relative motion between its different layers is called
- (a) surface Tension
 - (b) cohesion
 - (c) viscosity
 - (d) capillarity
- (iii) Which one of the following law is applicable to a hydraulic lift?
- (a) Kirchhoff's Law
 - (b) Pascal's Law
 - (c) Archimedes' Principle
 - (d) Archimedes' Law
- (iv) Continuity Equation is derived based on
- (a) conservation of mass principle
 - (b) conservation of momentum principle
 - (c) conservation of energy principle
 - (d) none of the above
- (v) The pressure above the atmospheric pressure is known as
- (a) suction pressure
 - (b) vacuum pressure
 - (c) gauge pressure
 - (d) none of the above
- (vi) Reynolds number is given as
- (a) Inertia Force / Viscous Force
 - (b) Buoyancy Force / Viscous Force
 - (c) Elastic Force / Viscous Force
 - (d) Pressure Force / Viscous Force
- (vii) Bernoulli's equation is derived from

- (a) Reynold's equation of Motion
 - (b) Navier-Stokes Equation
 - (c) Euler's equation of Motion
 - (d) None of the above
- (viii) If velocity potential exists in a fluid flow, then the flow is said to be
- (a) turbulent
 - (b) irrotational
 - (c) rotational
 - (d) laminar
- (ix) If m = total no. of variables and n = number of fundamental dimensions involved then number of π terms in Buckingham's π method would be
- (a) $m + n$
 - (b) $(m-n) + 1$
 - (c) $m \times n$
 - (d) $m - n$
- (x) The ratio of the square root of inertia force to surface tension force is known as
- (a) Froude number
 - (b) Euler number
 - (c) Weber Number
 - (d) Reynold's Number

Part B

Answer ANY FOUR questions

(4 x 5mark = 20 marks)

- Q.2 Define the following terms with respect to Fluid Mechanics: (i) specific weight; (ii) relative density; (iii) compressibility; (iv) surface tension; (v) steady and unsteady Flow. 1×5=5
- Q.3 State the Newton's law of viscosity with its mathematical expression. What are the different units available for absolute and kinematic viscosity? 2+3=5
- Q.4 A hydraulic press has a ram of 30 cm diameter and a plunger of 5 cm diameter. Find the weight lifted by the hydraulic press when the force applied at the plunger is 400 N. 5
- Q.5 Determine the total pressure and depth of centre of pressure on a plane rectangular surface of 1 m wide and 3 m deep when its upper edge is horizontal and (a) coincides with water surface; (b) 2 m below the free water surface. 5
- Q.6 A fluid flow field is given by $\vec{V} = x^2 y \hat{i} + y^2 z \hat{j} - (2xyz + yz^2) \hat{k}$. Prove that it is a case of possible steady incompressible fluid flow. 5
- Q.7 Water is flowing through a pipe of diameter 30 cm at a velocity of 4 m/s. Find the velocity of oil flowing in another pipe of diameter 10 cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water and oil is given as 0.01 poise and 0.025 poise. The sp.gr. of oil = 0.8. 5
- Q.8 Find the expression for the power 'T', developed by a pump when 'T' depends upon the Head 'G', the discharge 'P' and specific weight 'w' of the fluid. 5
- Q.9 A 25 cm diameter pipe carries oil of sp.gr. 0.9 at a velocity of 3 m/s. At another section the 5

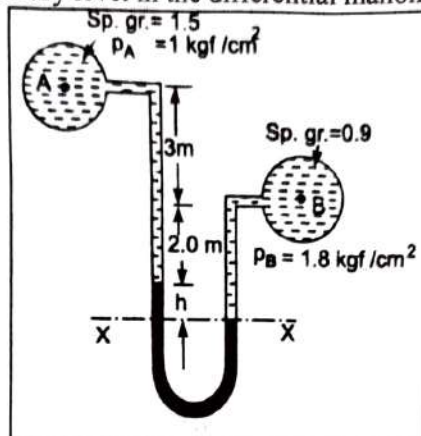
diameter is 20 cm. Find the velocity at this section and also mass rate of flow of oil.

Part C

Answer ANY TWO questions

(2 x 10mark = 20 marks)

- Q.10 The velocity vector in a fluid flow is given by $\vec{V} = 2x^3\hat{i} - 5x^2y\hat{j} + 4t\hat{k}$. Find the velocity and acceleration of a fluid particle at (1, 2, 3) at time $t=1$. 10
- Q.11 A differential manometer is connected at the two points A and B of two pipes as shown in the Figure. The pipe A contains a liquid of sp.gr. = 1.5 while pipe B contains a liquid of sp.gr. = 0.9. The pressure at A and B are 1 kgf/cm² and 1.80 kgf/cm² respectively. Find the difference in mercury level in the differential manometer. 10



- Q.12 The resisting force 'R' of a supersonic plane during flight can be considered as dependent upon the length of aircraft 'l', velocity 'v', air viscosity ' μ ', air density ' ρ ' and the bulk modulus of air 'K'. Express the functional relationship between these variables and the resisting force. 10
- Q.13 A ship 300 m long moves in sea water, whose density is 1030 kg/m³. A 1:100 model of this ship is to be tested in a wind tunnel. The velocity of air in the wind tunnel around the model is 30 m/s and the resistance of the model is 60 N. Determine the velocity of the ship in sea water and also the resistance of the ship in sea water. The density of air is given as 1.24 kg/m³. Take $\nu_{sea\ water} = 0.012$ stokes and $\nu_{air} = 0.018$ stokes. 10